

demo dashboard ui

1. Objective

The dashboard must allow a viewer to:

- observe system behavior over time
- compare phases (Phase 1 → Phase 5)
- understand why a decision was made
- evaluate robustness under attack

2. Layout Overview

Main Structure

Header (controls + scenario selector)
Left Panel Center Panel Right Panel

| Bottom Panel (metrics + charts) |

3. Header (Global Controls)

Controls

Scenario Selector: - Clean - Noisy - Coordinated Attack - Sybil Attack - Location Spoofing Phase Selector: - M1 → M5 Simulation Controls: - Play / Pause - Speed slider - Reset Filters: - show only suspicious users - show only high-impact posts

4. Left Panel — User Intelligence

User Table

Columns: user_id R_i (reliability) Exp_i Anom_i T_i (if Phase 3+) weight w_i

Visual Encoding

color: green → trusted red → suspicious size: activity level

Interaction

click user → show: - activity timeline - anomaly breakdown - coordination links

5. Center Panel — Network Graph (Core Visual)

Graph View

Nodes = users Edges = similarity (A_{ij})

Node Encoding

size → weight w_i color → anomaly score border → trusted (T_i high)

Edge Encoding

thickness → similarity strength

Dynamic Behavior

- graph evolves over time
- clusters form during attacks

Key Demo Moment

Switch Phase 2 → Phase 3 Observe: coordinated cluster becomes visible trust redistribution happens

6. Right Panel — Post Intelligence

Post List

Columns: post_id $C_j(t)$ $Var_j(t)$ $N_j(t)$ status (trusted / uncertain / rejected)

Selected Post Detail

When clicked:

show: Credibility Breakdown: C_{Bayes} (C_{ML} , C_{memory} if Phase 5) Signal Breakdown: S_{j+} , S_{j-} -contributing users Temporal Graph: $C_j(t)$ over time Alert Status: triggered / not triggered

7. Bottom Panel — Metrics & Evaluation

Tabs

Tab 1: Accuracy Metrics

Accuracy over time Brier score

Tab 2: Attack Analysis

attack success rate false positives false negatives

Tab 3: Alert Quality

precision recall alerts triggered over time

Tab 4: System Comparison

bar chart: M1 vs M2 vs M3 vs M4 vs M5

8. Key Visualizations

9. Credibility Timeline

x-axis → time y-axis → $C_j(t)$ plot: true posts (green) false posts (red)

2. Anomaly Distribution

histogram of Anom_i

3. Trust Propagation Effect

before vs after T_i update

4. Coordination Detection

cluster highlight: bot group visually grouped

9. Demo Scenarios (Scripted Flow)

Demo 1: Clean System

show: stable credibility no anomalies

Demo 2: Coordinated Attack

Phase 2: system fooled Phase 3: cluster detected weights drop credibility corrected

Demo 3: Location Relevance

Phase 4: alerts only to nearby users

Demo 4: Early Detection

Phase 5: ML predicts credibility early

10. Tech Stack Recommendation

Frontend

React + Next.js

Visualization

D3.js (graph) Recharts / Chart.js (plots)

State Management

Redux

Backend (for demo)

FastAPI WebSocket for real-time updates

11. Data API Contract

Fetch State

GET /simulation/state

Response

```
{ "users": [...], "posts": [...], "edges": [...], "metrics": {...} }
```

Control Simulation

```
POST /simulation/control { "action": "play | pause | reset", "scenario": "attack", "phase": "M3" }
```

12. UX Principles (Critical)

13. Show Cause, Not Just Output

Bad:

$C_j = 0.82$

Good:

$C_j = 0.82$ because: trusted users $\uparrow$ anomaly $\downarrow$

2. Make Phase Differences Obvious

toggle M2 vs M3 $\rightarrow$ instant visual change

3. Highlight Failures

show where system fails in early phases

Fix: Phase-Specific Dashboard Extensions

Phase 3 — Network Intelligence Panel (Required Additions)

Add: Trust Propagation View

Panel: Trust Flow Show: R_i (base reliability) $\rightarrow$ T_i (after propagation) Visualization: bar or node overlay: before vs after trust

Add: Coordination Cluster Detection

Highlight: clusters where $S_{coord} > threshold$ UI: draw bounding box or color cluster

Add: Coordination Inspector (on click)

Selected user i : Show: top similar users j $Sim(i,j)$ time alignment

Critical Toggle

Toggle: "Show coordination influence" Effect: remove coordination term $\rightarrow$ observe failure

Phase 4 — Spatial Intelligence Panel (Required Additions)

Add: Map Visualization (Mandatory)

Map: users $\rightarrow$ points posts $\rightarrow$ markers

Encoding

User: color → L_i (location confidence) radius → influence Post: circle radius → propagation radius r_j

Add: Proximity Visualization

For selected user u : draw: heatmap of $\text{Prox}(u,j,t)$

Add: Location Confidence Breakdown

For user i : show: S_{gps} S_{speed} S_{source} S_{cont} → L_i

Add: Alert Filtering Explanation

For each alert: show: $\text{Prox}(u,j,t)$ threshold τ_p decision = pass/fail

Phase 5 — ML Intelligence Panel (Required Additions)

Add: Credibility Decomposition

$C_j = X\% C_{\text{Bayes}} Y\% C_{\text{ML}} Z\% C_{\text{memory}}$

Add: ML Feature Importance

For post j : Top features: keyword_score velocity early_votes

(Use simple feature weights or SHAP-style bars)

Add: Memory Neighbors View

Show: similar posts k Display: similarity score their credibility

Add: ML vs Rule Comparison Toggle

Toggle: "Disable ML" Observe: early-stage failure vs success

Add: Anomaly ML vs Rule Split

For user i : Anom_rule Anom_ML Final Anom_i

Global Enhancement (All Phases)

Phase Comparison Mode (Very Important)

Split screen: Left → Phase N Right → Phase $N+1$ Compare: credibility alerts anomalies

Decision Trace Panel (Core Requirement)

For ANY decision:

Why was this alert triggered? Show chain: User signals → w_i → Post signals → C_j → Proximity → Prox → Final decision

What This Fix Achieves

Before

Dashboard shows results

After

Dashboard explains WHY results differ across phases